

EXTRA TASK

Exercise A:

1. Draw the conceptual model
2. Identify the possible Entities, their Attributes and the relationships between the entities from the given scenario.
3. Draw the ERD following diagramming convention.
4. Draw the physical model
5. Write the sentences for each relationship, in both directions.
6. Map the relationships.
7. Draw the matrix diagram.

We need to store data about each of our company's employees. We need to track each employee's first name, last name, job or position, hire date and salary, office telephone number. The employees can be involved in more projects that are carried out within the company. The name of each project is known, the date on which the project started and it is possible to know a date when the project will be completed. Only one employee works on each project, but one employee can be involved in several projects. Each employee is assigned a unique employee number. Our company is divided into departments. Each employee reports to a department -- for example, accounting, sales, or development. We need to know the department location for each employee. Each department has a unique number. Each department has several offices, located in one of the company's buildings. An office may have many printers but there may not be any printers in some offices.

Exercise B: Draw softboxes and relationship lines for each relationship in both directions.

- a) Each EMPLOYEE must be assigned to one and only one DEPARTMENT. Each DEPARTMENT must be responsible for one or more EMPLOYEES.
- b) Each EMPLOYEE must be assigned to exactly one DEPARTMENT. Each DEPARTMENT may be responsible for exactly one EMPLOYEE.
- c) Each TOWN may be the birthplace of one or more people. Each PERSON must be born in one and only one TOWN.
- d) Each PERSON may be visitor of one or more TOWNS. Each TOWN must visit by many PEOPLE.

Exercise C: Determine how exclusive relationships should be modeled in the following scenario.

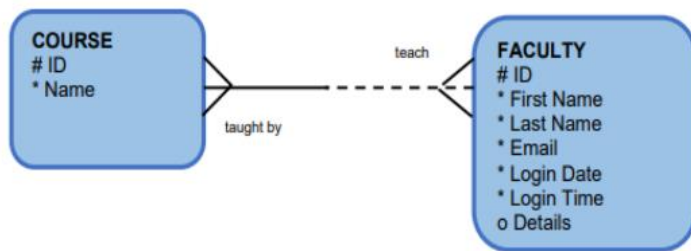
Each COURSE instance in the Academic Database can either be held ONLINE or in a SEATED location. Each SEATED location has a building name, room number and a date/time when the COURSE is offered. The ONLINE classes have a logon id and a password required to enter the COURSE.

Exercise D: Develop a hierarchical structure to represent the following situation.

Curves Dynamics sells products throughout the United States. They are divided into four major sales regions: the Northern, Eastern, Southern, and Western regions. Each

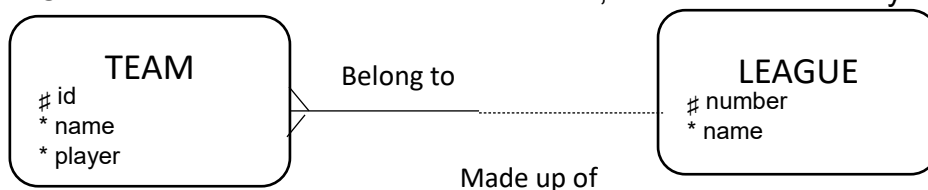
sales region is then divided into sales districts. Each district is made up of sales territories.

Exercise E: Resolve M: M relationships between FACULTY and the COURSE.

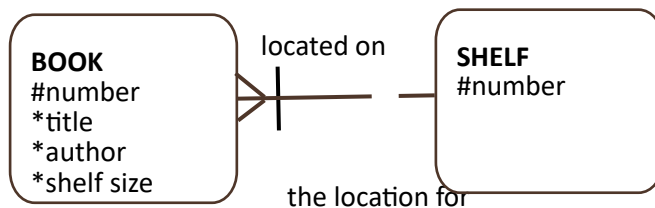


Exercise F:

1. Check to see if the ERD is in 1NF. If not, make the necessary changes to correct it.



2. The UID of a library book includes its BOOK number and SHELF number. Does this ERD follow the rules of Second Normal Form? If you spot a violation, correct it.



3. Cass Enrollment is the intersection entity that resolves the M:M between STUDENT and CLASS. Does the ERD follow the rules of Second Normal Form? If you spot a violation, correct it.

